

Roman set square
c.1st century BCE

This bronze instrument would have been important to Roman builders, helping them set construction blocks exactly square.

straight edges
set in a right
angle

Laser spirit level
21st century

An instrument used in construction for measuring vertical angles, this level defines a level plane along the beam of a laser.



rotating base
fixed on
tripod



Circumferentor
1676

Used by surveyors before the invention of the theodolite, the circumferentor measured angles and could be used horizontally and vertically to calculate distances.

complete circle
scale or brass ring

Brass half-circle theodolite
19th century

A theodolite measures horizontal and vertical angles and is an important tool in surveying. The instrument's telescope is focused on a distant object, the position of which is determined with respect to horizontal and vertical scales.



eye-piece

vertical
half-circle
graduated
into degrees

leveling
plate

Nesting cups
19th century

Cuplike standard weights used with counterbalancing mechanical scales could be nested together in multiples.



pound

Standard weights
19th century

Many countries today have replaced the pound with the kilogram.



Lead weight
c.250 BCE

Greek merchants used standard weights—usually made from lead, and fashioned into rectangles.



Jade weight
Date unknown

In early Chinese civilizations, precious minerals such as jade were used for standard weights.

Grain measure
Traditional

Fixed quantities of grains, such as wheat or barley, were once used as standard units of mass.



Graduated pipette
21st century

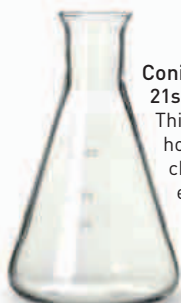
Glass pipettes, graduated in fractions of milliliters, can measure liquid volumes precisely, drop by drop.



milliliter
graduations

Conical glass flask
21st century

This flask is used as a holding container for chemical reactions in experiments where total volume does not have to be accurate.



kilogram



MEASURING INSTRUMENTS

SIMPLE OR SOPHISTICATED, THERE ARE MEASURING INSTRUMENTS FOR ALL PURPOSES

In everyday life, exact measurement is not always important. A wooden cup can be adequate for delivering a fair share of grain, but scientists who want to know the dimensions of microscopic objects need to use precision instruments.

In scientific experiments or studies, measurements must be made with an appropriate level of care and accuracy to ensure that results and conclusions are reliable. Scientists require their measuring instruments to give values within an acceptable margin of error, using standard units that are recognized universally. Today, nearly all countries use the *Système Internationale (SI)*—the modern form of the metric system—which was introduced in the 1960s.